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Surface-engineered Ni–Pt alloys as robust and cost-effective electrocatalysts for high-performance proton exchange membrane water electrolysis

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The development of efficient and durable electrocatalysts for the hydrogen evolution reaction (HER) is essential to advance proton exchange membrane water electrolysis (PEMWE), a key technology for green hydrogen production. Although Pt remains the benchmark HER catalyst, its high cost and limited durability under harsh acidic conditions hinder large-scale deployment. To address these challenges, we developed a Ni–Pt alloy catalyst *via* scalable electrodeposition followed by thermal annealing. Structural characterizations confirmed that thermal treatment induced Pt surface segregation, yielding a Pt-enriched shell with an Ni-rich core. This thermally induced surface reconstruction markedly reduced both surface and cohesive energies relative to pure Pt, thereby enhancing thermodynamic and structural stability. Density functional theory (DFT) calculations revealed that when Pt surface coverage approached ~50%, the alloy surface exhibited a hydrogen adsorption free energy (ΔE_{H*}) close to the thermodynamic optimum. In HER testing, the heat-treated $Ni_{98.7}Pt_{1.3}$ catalyst (HT-$Ni_{98.7}Pt_{1.3}$) achieved a low overpotential of just 14 mV at −10 mA cm^{-2}, while maintaining robust stability over 5000 accelerated durability cycles. Moreover, when integrated into a PEMWE cell, the HT-$Ni_{98.7}Pt_{1.3}$ electrode delivered a record-high mass activity of 27 A mg_{Pt}^{-1} at 1.7 V and sustained performance for over 200 h at 1 A cm^{-2}, surpassing commercial Pt/C. The synergy of scalable synthesis, structural robustness, and outstanding catalytic activity highlights surface-engineered Ni–Pt alloys as highly promising next-generation catalysts for sustainable hydrogen production.

Introduction

The development of efficient and durable electrocatalysts is pivotal for advancing sustainable energy conversion technologies.[1,2] Among various candidates, Pt is regarded as the benchmark catalyst for the hydrogen evolution reaction (HER), whereas Ir- and Ru-based materials are widely recognized as the most effective catalysts for the oxygen evolution reaction (OER) in proton exchange membrane water electrolysis (PEMWE) systems, owing to its exceptional catalytic activity and chemical stability.[1,3–5] However, the high cost, limited reserves, and insufficient long-term stability of Pt under harsh electrochemical conditions present substantial challenges to its large-scale commercialization. In response, considerable research efforts have focused on non-precious metal catalysts such as Ni, Co, Cu, and Mo-based materials for HER.[6–9] Although these alternatives offer clear cost benefits, their catalytic activity and durability in acidic media remain inadequate for practical application. As such, there is a pressing need for Pt-based catalysts that can minimize Pt loading while maintaining, or even enhancing, catalytic activity and durability.[10]

Electrodes for membrane electrode assemblies (MEAs) are typically fabricated by dispersing nanoparticulate catalysts onto carbon-based supports (*e.g.*, carbon black, carbon nanotubes), bound together with polymeric ionomers or binders such as Nafion.[3] This catalyst-coated electrode configuration remains the most widely adopted architecture in commercial PEMWE systems due to its well-established scalability, relatively high catalyst utilization, and effective ionic transport pathways provided by the ionomer network. However, weak catalyst–support interactions often lead to catalyst agglomeration, detachment, or dissolution under harsh electrochemical environments.[11,12] Moreover, polymeric binders, though essential for structural integrity and ionic conductivity, can block active catalytic sites and introduce additional interfacial resistance at

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the catalyst–support and catalyst–electrolyte interfaces.[13,14] Thus, while this architecture offers clear advantages, it inherently involves a trade-off between ionic connectivity and accessible catalytic surface area. These drawbacks become more pronounced under dynamic operating conditions (*e.g.*, frequent start-up/shut-down cycles or prolonged high-current operation), ultimately resulting in catalyst degradation, performance loss, and reduced MEA durability.[11]

To overcome these challenges, self-supported catalyst electrodes have emerged as a promising alternative. These electrodes, constructed without binders or external supports, integrate directly with the porous transport layer, thereby minimizing interfacial resistance and maximizing the electrochemically active surface area.[13,15,16] In addition, the self-supported architecture provides superior mechanical robustness and more efficient mass transport pathways, enhancing catalytic performance and operational stability.

Non-precious group metals (non-PGMs) such as Ni, Co, and Fe are not only cost-effective but also possess smaller atomic radii compared to Pt, enabling favorable electronic modifications when alloyed.[17] Their incorporation increases the electron density of Pt and subtly shifts the d-band center away from the Fermi level,[18,19] thereby weakening hydrogen adsorption and bringing it closer to the thermodynamic optimum for HER.[20] As a result, Pt-based alloys containing small amounts of non-PGMs have been extensively investigated, with nanostructures such as nanoparticles and nanowires widely synthesized and evaluated.[21–24] In addition, recent studies have demonstrated that multicomponent high-entropy alloys can further enhance catalytic performance by exploiting synergistic interactions among multiple elements, achieving remarkable high mass- and specific activities even with reduced noble metal contents.[25–27] These advances highlight the effectiveness of compositional and electronic structure engineering in improving HER activity. However, a large portion of studies remain focused on Pt-rich compositions, which still suffer from high precious metal content and limited economic viability. In contrast, catalysts with a predominantly non-PGM bulk composition (*e.g.*, Ni-rich alloys) can significantly reduce material costs, but they generally exhibit much lower intrinsic HER activity at the surface. Therefore, an ideal catalyst design should decouple bulk composition from surface structure by employing a cost-effective Ni-rich bulk matrix while selectively creating a highly active Pt-enriched surface (*e.g.*, a core@shell-like structure).[28]

To date, however, most surface engineering strategies based on Pt segregation have been limited to powder-type nanoparticle systems, and their translation into binder-free, self-supported electrode architectures capable of sustaining industrial-level PEMWE operation remains insufficiently explored. In this context, the key conceptual advance of this work is the direct integration of thermodynamically driven Pt surface segregation with a scalable, binder-free three-dimensional electrode architecture, enabling simultaneous control of atomic-scale surface chemistry and macroscopic electrode design within a unified platform.

Given these considerations, a scalable synthesis route that enables the fabrication of non-PGM-rich Pt alloys with engineered surface structures is highly desirable. Electrodeposition offers several advantages, including operation under ambient conditions, reduced capital and operational costs, and excellent conformity on complex geometries. More importantly, it allows precise control over the alloy composition, morphology, crystallinity, and thickness by tuning parameters such as deposition solution composition, deposition potential, and time.[29] These merits make electrodeposition a highly attractive strategy for fabricating composition-tunable, application-specific electrocatalyst architectures.

This work aims to develop a cost-effective and scalable strategy for engineering self-supported, Pt-efficient electrocatalysts. To this end, we prepared a self-supported Ni-rich Ni–Pt electrode *via* electrodeposition followed by thermal annealing to induce surface restructuring. Upon annealing, the Ni-rich Ni–Pt electrode undergoes thermally driven surface reconstruction, whereby Pt atoms segregate to the surface due to surface energy differences and strong interaction with hydrogen, resulting in a Pt-enriched surface layer.[30] This unique surface configuration enhances the intrinsic HER activity and improves durability under acidic conditions. Density functional theory (DFT) calculations further reveal that the reconstructed Ni–Pt surface possesses lower surface and cohesive energies compared to pure Pt, indicating superior thermodynamic and structural stability. Moreover, the Pt-enriched surface facilitates H–H bond formation by reducing the associated free energy barrier, thereby optimizing the reaction pathway. Electrochemical evaluation confirms that the heat-treated Ni–Pt electrode, despite containing only trace amounts of Pt, achieved remarkable HER performance and long-term stability. When integrated into a PEMWE cell as the cathode, the electrode demonstrates outstanding performance and operational durability, underscoring its potential for practical application. The findings provide insights into the rational design of durable HER electrodes, offering promising prospects for next-generation PEMWE systems.

Experimental section

Catalyst synthesis *via* electrodeposition

Ni, Pt, and Ni–Pt catalysts were synthesized using a controlled electrodeposition method in a custom-designed three-electrode electrochemical cell. A potentiostat (Metrohm Autolab, PGSTAT302N) was used to precisely regulate the electrochemical conditions. For the three-electrode configuration, carbon paper (CP) (Ballard, Avcarb MGL280) was selected as the working electrode, whereas a graphite rod (WonATech, GR002H) and a saturated calomel electrode (SCE) (CHI Instruments, CHI150) were used as the counter and reference electrodes. To enhance the surface hydrophilicity of the CP substrate, it was subjected to a pre-treatment process involving sonication in a nitric acid (HNO_3; Junsei, 37335-1250) solution for 15 min.[31,32] The electrodeposition bath was prepared by dissolving nickel chloride hexahydrate ($NiCl_2 \cdot 6H_2O$; Wako Pure Chemical Industries, 141-01045) and chloroplatinic acid hexahydrate ($H_2PtCl_6 \cdot 6H_2O$; Sigma-Aldrich, 206083) in deionized (DI; 18.2 MΩ) water, serving as the Ni and Pt precursors, respectively. To tune the electrolyte properties, hydrochloric acid (HCl; Daejung, 4090-4400), ammonium chloride (NH_4Cl; Daejung, 1060-4100), and boric acid (H_3BO_3; Daejung 2036-4405) were added. The buffering action of H_3BO_3 was essential for sustaining electrolyte pH stability during electrodeposition, particularly under concurrent hydrogen evolution, which would otherwise promote the undesirable metal hydroxide precipitation.[33]

Prior to deposition, the electrolyte solution was purged with N_2 gas for 30 min to remove dissolved oxygen. Ni_{100}, Pt_{100}, and Ni–Pt electrodes were fabricated by applying a constant potential of -3.0 V_{SCE} to the CP working electrode for 10 min. The geometrically exposed area of CP in contact with the electrolyte was 1.327 cm^2. After electrodeposition, the $Ni_{98.7}Pt_{1.3}$ pre-electrode (P–$Ni_{98.7}Pt_{1.3}$) underwent heat treatment at 200 °C for 2 h in an Ar atmosphere containing 4% H_2, with the temperature increased at a rate of 10 °C min^{-1},[34,35] yielding the heat-treated $Ni_{98.7}Pt_{1.3}$ sample (HT-$Ni_{98.7}Pt_{1.3}$). The electrodeposition parameters employed in this study are listed in Table S1, and the current density-time profiles are provided in Fig. S1.

Material characterizations

The surface morphologies of the synthesized electrodes were examined using field emission scanning electron microscopy (FE-SEM; Sigma, Carl Zeiss). The bulk atomic compositions of the prepared materials, along with the quantification of elemental loading, were determined using energy dispersive X-ray spectroscopy (EDX; Thermo Fisher, Noran System 7), inductively coupled plasma optical emission spectroscopy (ICP-OES; Agilent Technologies, Model 5900), and inductively coupled plasma mass spectrometry (ICP-MS; Agilent Technologies, Model 7900). Elemental distributions were further analyzed using a transmission electron microscope (TEM; Thermo Fisher Scientific, Talos F200X-D6328). To prepare TEM specimens, portions of the electrodeposited layer were dispersed into isopropanol using sonication and subsequently dropped onto TEM grids for imaging and elemental mapping. Crystalline structures were characterized using X-ray diffraction (XRD, Bruker-AXS, New D8-Advance). XRD patterns were collected at a scan rate of 2° min^{-1} over a two-theta range of 20−80°. To investigate the electronic structure and surface composition of the catalyst, X-ray photoelectron spectroscopy (XPS, Thermo Fisher Scientific, K-alpha+) was employed. XPS binding energies were calibrated by assigning the C–C component of the C 1s peak to 284.8 eV.

Computational methods

DFT calculations were performed to investigate the enhanced HER performance on Pt-alloyed Ni surfaces using the Vienna *ab initio* simulation package.[36] The core and valence electrons were described using projected-augmented wave (PAW) potentials[37] and a plane-wave basis set[38] with a cutoff energy of 520 eV. The optB86b-vdW exchange-correlation functional was employed to accurately account for van der Waals interactions.[39] All calculations were performed with spin-polarization taken into account. A gamma-centered *k*-point sampling method with a 10 × 10 × 1 mesh was used for the slab models. All calculations were relaxed to an electronic convergence criterion of 10^{-4} eV and atomic force tolerance of 0.03 eV $Å^{-1}$.

A five-layer Ni_3Pt_1 (111) surface slab model was constructed with a vacuum layer exceeding 20 Å to prevent interactions between periodic images. Each surface layer contained three Ni atoms and one Pt atom, forming a Ni_3Pt_1 (111)-$(2\sqrt{3} \times 2\sqrt{3})$ R30° structure. During the surface model relaxation, the two lowest atomic layers were constrained, while the top three layers are allowed to relax. In hydrogen-adsorbed models, only the hydrogen atom was allowed to relax. Pt atoms were distributed at the surface with coverages of 1/4, 2/4, 3/4, and 1 monolayer (ML). For comparison, a five-layer Pt (111) slab with an FCC structure and an equivalent $(2\sqrt{3} \times 2\sqrt{3})$ R30° supercell was also constructed under identical setting.

The surface energy was evaluated as follows:

$$E_{\text{surf}} = \frac{1}{2A}(E_{\text{slab}} - N_{\text{Ni}}\mu_{\text{Ni}} - N_{\text{Pt}}\mu_{\text{Pt}}), \quad (1)$$

where E_{slab} is the total energy of the slab model, N_{Ni} and N_{Pt} are the numbers of Ni and Pt atoms in the slab model. μ_{Ni} and μ_{Pt} represent the chemical potentials of Ni and Pt, referenced to the per-atom energies of their respective bulk phases. The cohesive (or binding) energy of a surface Pt atom, $E_{\text{coh.}}$, was defined as follows:

$$E_{\text{coh.}} = E_{\text{slab}} - E_{\text{slab-Pt}} - \mu_{\text{Pt}}, \quad (2)$$

where $E_{\text{slab-Pt}}$ is the total energy of the same slab after removing one Pt atom from the top layer.

The adsorption energy of hydrogen on the surface ($E_{\text{H}*}$) was calculated as follows:

$$E_{\text{H}*} = E_{\text{slab+H}*} - E_{\text{slab}} - E_{\text{H}_2}/2, \quad (3)$$

where $E_{\text{slab+H}*}$ and E_{slab} denote the total energies of the surface with and without hydrogen adsorption, respectively.$E_{\text{H}_2}/2$ corresponds to half of the energy of molecular hydrogen in the gas phase, derived from the computational hydrogen electrode approach.[40]

Electrochemical measurements

Cyclic voltammetry (CV) measurements for HER assessment were conducted in 0.5 M H_2SO_4 under a three-electrode configuration, employing an SCE and a graphite rod as the reference and counter electrodes, respectively. The use of a graphite rod instead of Pt was critical to avoid Pt dissolution and subsequent redeposition on the working electrode, which could otherwise interfere with the HER signal.[41] CV scans were conducted at 2 mV s^{-1} over a potential window from 0.02 to -0.33 V_{RHE}.

Accelerated degradation tests (ADT) were performed by subjecting the catalysts to 5000 continuous potential cycles at a scan rate of 100 mV s^{-1} within the same HER potential window. Polarization data were referenced to the reversible hydrogen electrode (RHE). Ohmic losses were corrected based on resistance values obtained from electrochemical impedance spectroscopy (EIS) (Wuhan Corrtest Instruments Corp. Ltd, CS310). Prior to electrochemical testing, the electrolyte was deaerated by purging with N_2 for 30 min to eliminate dissolved oxygen.

PEMWE single cell evaluation

The cathode of the MEA was fabricated using the Ni–Pt/CP electrode identified as optimal in half-cell HER evaluations, without applying any additional post-processing steps, including ionomer/binder coating or hot-pressing, thereby highlighting its self-supported configuration. For the anode, a commercial IrO_x catalyst was spray-coated onto CP or the Ti porous transport layer (PTL) (Ti fiber felt; Bekaert 2GDL08N-025) at a loading of 2.0 mg_{Ir} cm^{-2}.

For anode ink preparation, IrO_x catalyst (Alfa Aesar, 043396) was dispersed in a mixture of DI water, isopropanol, and Nafion® D-520 at a fixed mass ratio (9 : 50 : 236 : 20), while commercially available Pt/C cathodes were prepared by spray-coating onto CP with a Pt loading of 0.1 mg_{Pt} cm^{-2}. The ink for Pt/C catalyst was formulated by dispersing Pt/C catalyst (Alfa Aesar, 47308) in a solvent mixture of DI water, isopropanol, and Nafion® D-520, using a mass ratio of 5 : 50 : 157 : 36. Nafion® 212 with a thickness of 50.8 μm served as the proton exchange membrane. The MEA active area was standardized to 1×1 cm^2.

PEM water electrolysis was conducted at a cell temperature of 90 °C. Preheated DI water was fed only to the anode at a flow rate of 15 mL min^{-1} and its temperature was controlled using a 95 °C line heater. Following a 20 min initial activation step at 2.3 V_{cell}, the electrolyzer performance was assessed by stepping the cell voltages from 2.3 down to 1.35 V_{cell} with 50 mV interval, holding each voltage for 1 min. Long-term stability was assessed under continuous operation for 200 h at a current density of 1 A cm^{-2}.

Results and discussion

Synthesis and characterization of Ni–Pt electrodes

Fig. 1a schematically illustrates the preparation of the Ni–Pt catalyst-integrated porous transport electrode. To improve surface wettability prior to electrodeposition, the CP was first sonicated in HNO_3 solution to introduce oxygen-containing functional groups, primarily carbonyl and carboxyl groups.[31,32] After rinsing with DI water, the pre-electrode was synthesized *via* electrodeposition. The electrodeposition conditions and bulk compositions of various samples measured by EDX are presented in Table S1. For a more precise determination of the bulk composition, ICP-OES analysis was conducted, yielding 98.5 at% Ni and 1.5 at% Pt. These values showed excellent agreement with the EDX results (98.7 at% Ni and 1.3 at% Pt), indicating that the EDX-based compositional labeling provides a reliable estimate of the alloy composition within its semi-quantitative limits. Following electrodeposition, the electrodes were subjected to thermal treatment at 200 °C for 2 h under a 4% H_2/Ar atmosphere to promote surface stabilization. The surface morphologies of the representative samples of as-prepared P–$Ni_{98.7}Pt_{1.3}$ and HT-$Ni_{98.7}Pt_{1.3}$ catalysts were examined using FE-SEM (Fig. 1b and c). For comparison, the morphologies of bare CP and other compositions (without heat-treatment) are shown in Fig. S2. The carbon fibers of the pristine CP exhibited a smooth surface (Fig. S2a), whereas the Ni_{100} catalyst (Fig. S2b) displayed a dendritic deposition morphology with a rough surface. Such a dendritic structure arises from electrodeposition under mass transport-controlled conditions at −3.0 V_{SCE}, accompanied by vigorous hydrogen evolution. The slow release of trapped hydrogen during deposition led to localized electrode growth around gas bubbles, producing irregular features (Fig. S2b inset).[42] In contrast, Pt_{100} (Fig. S2f) exhibited very low deposition efficiency due to the low concentration of the Pt precursor, resulting in the sparse, uniform deposition of small Pt nanoparticles across the fibers. Further, Ni–Pt alloy catalysts (Fig. S2c–e) consistently showed more uniform deposits composed of rounded grains, regardless of composition. Notably, heat treatment had little influence on the overall morphology of the $Ni_{98.7}Pt_{1.3}$ catalyst (Fig. 1b and c).

The elemental distribution of Ni and Pt in P–$Ni_{98.7}Pt_{1.3}$ and HT-$Ni_{98.7}Pt_{1.3}$ was analyzed using TEM elemental mapping (Fig. 1d and e). It should be noted that TEM analysis of electrodeposited samples is challenging due to the nature of the as-deposited films. Although sonication was used to preferentially detach surface portion, the collected specimen may include both surface and subsurface material, and thus do not exclusively represent the pristine electrode surface. Quantitative analysis of the TEM-mapped specimen revealed a composition of approximately 79.8 at% Ni and 20.2 at% Pt, which is considerably different from the bulk composition determined by EDX (∼98.7 at% Ni and 1.3 at% Pt). This discrepancy indicates that the analyzed portion primarily reflect the surface composition of the electrodeposited layer. Such differences between bulk and surface composition are further corroborated and discussed in the XPS analysis presented below. Taken together, these findings suggest that the electrodeposited layer exhibits Pt enrichment at the surface. This preferential surface segregation of Pt is attributed to its larger atomic radius,[43] which drives its migration toward the surface. Upon thermal annealing under H_2/Ar, more pronounced segregation was observed; HT-$Ni_{98.7}Pt_{1.3}$ exhibited a core-shell-like structure, with Pt concentrated at the surfaces. Quantitative mapping confirmed a composition of ∼57.2 at% Ni and 42.8 at% Pt, highlighting the formation of a Pt-enriched surface layer. This segregation is driven by the strong Pt–H_2 interaction under annealing conditions[30] and the larger atomic radius of Pt.[43]

The crystal structures of the prepared catalysts were examined using XRD, and the corresponding patterns are shown in Fig. 1f (two-theta range of 35–60°) and Fig. S3 (two-theta range of 20–80°), alongside the reference pattern for the CP substrate. XRD patterns of CP exhibited characteristic peaks at 26.6° (004), 43.02° (101), 44.67° (102), and 54.79° (008) of carbon (JCPDS #26-1080). In addition to these peaks, the Ni_{100} sample displayed distinct diffraction peaks at approximately 44.51°, 51.85°, and 76.4°, consistent with a face-centered cubic structure of Ni (JCPDS #04-0850). With the addition of small

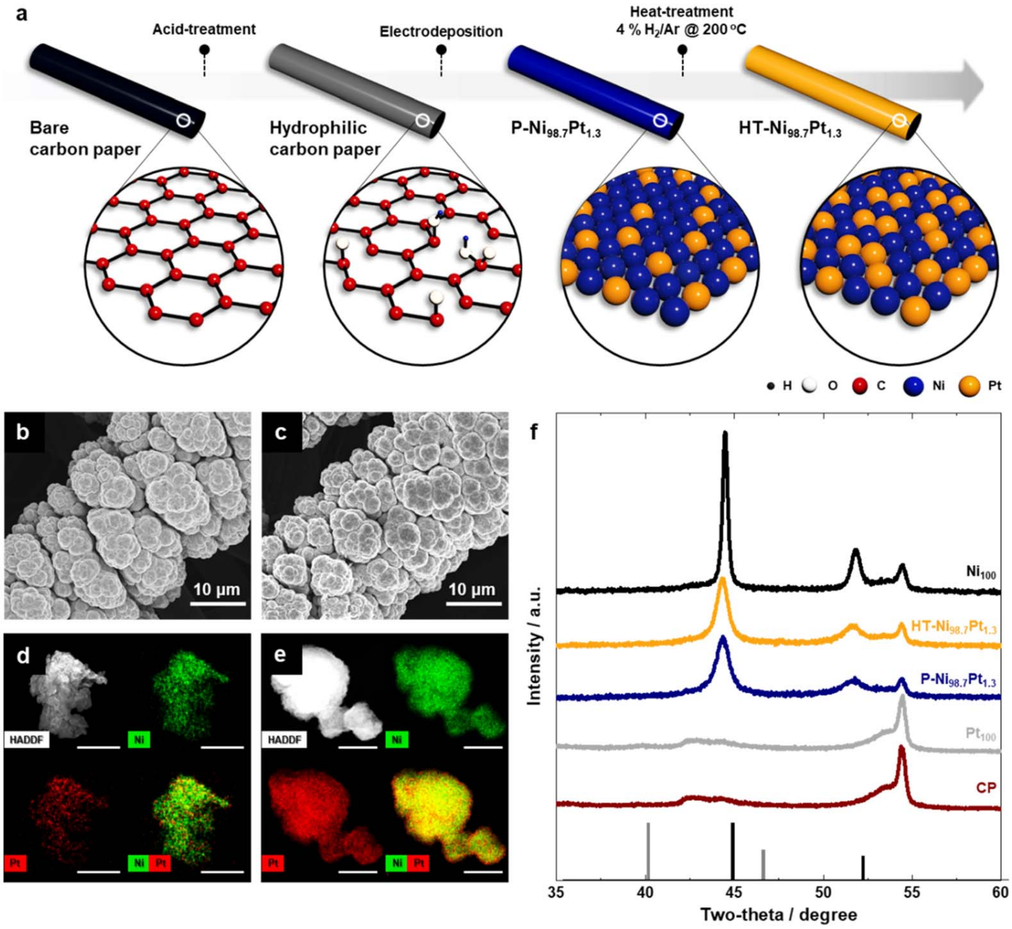

Fig. 1 (a) Schematic illustration of the preparation procedure of the Ni–Pt catalyst. FE-SEM images of (b) P–$Ni_{98.7}Pt_{1.3}$ and (c) HT-$Ni_{98.7}Pt_{1.3}$ samples electrodeposited on carbon paper (scale bar: 10 μm). TEM elemental mapping of Ni and Pt for (d) P–$Ni_{98.7}Pt_{1.3}$ and (e) HT-$Ni_{98.7}Pt_{1.3}$, showing the dispersion of Ni (green) and Pt (red), respectively (scale bar: 50 nm). (f) XRD patterns of bare carbon paper and electrodeposited Ni_{100}, P–$Ni_{98.7}Pt_{1.3}$, Pt_{100}, and HT-$Ni_{98.7}Pt_{1.3}$ samples; Solid markers at the bottom indicate reference patterns of Ni(04-0850, black) and Pt(04-0802, gray).

amounts of Pt, the Ni peaks slightly shifted to lower angles, indicative of Ni–Pt alloy formation. Concurrently, the diffraction peaks became weaker while exhibiting noticeable broadening of full width at half maximum (FWHM), consistent with a reduction in crystallite size from 26.1 to 13.2 nm. Notably, no distinct Pt peaks were observed in any Ni–Pt samples, regardless of Pt content (Fig. S3). Similarly, the Pt_{100} sample showed no detectable peaks apart from those of the CP substrate, suggesting that Pt was present in either a nanocrystalline or an amorphous state. Following annealing of $Ni_{98.7}Pt_{1.3}$, the diffraction peaks of HT-$Ni_{98.7}Pt_{1.3}$ exhibited a modest increase in intensity and a narrowing of FWHM, which can be attributed to thermally driven grain growth (14.0 nm) *via* particle aggregation.

The electronic structure and surface composition of Ni_{100}, Pt_{100}, and $Ni_{98.7}Pt_{1.3}$ electrodes before and after heat treatment were analyzed using XPS (Fig. 2 and Table S2). For Ni_{100} and Pt_{100}, both metallic and oxidized species were identified. In the case of $Ni_{98.7}Pt_{1.3}$, incorporation of a small fraction of Pt into the Ni matrix induced electron transfer from Ni to Pt, leading to a reduced portion of metallic Ni. Consistently, all Ni species exhibited a slightly higher binding energy, while Pt shifted to a lower binding energy, indicating an overall electron-deficient state of Ni and an electron-rich state of Pt. After annealing under a 4% H_2/Ar atmosphere, Ni oxide/hydroxide species were substantially reduced, resulting in a pronounced increase in metallic Ni. However, despite this reduction, the Ni 2p peaks shifted further to higher binding energies, while the Pt 4f peaks shifted further to lower binding energies. This behavior is attributed to enhanced Pt–Pt connectivity and increased electron delocalization, likely driven by Pt surface enrichment during annealing.

The correlation between electronic and structural rearrangement was further examined through quantitative surface analysis. XPS data (Table S2), based on Ni $2p_{3/2}$ and Pt 4f peaks with their respective thermo-modified Scofield sensitivity factors, revealed pronounced Pt surface enrichment. While the bulk composition of $Ni_{98.7}Pt_{1.3}$ measured by EDX was 98.7 at% Ni and 1.3 at% Pt, the surface composition measured by XPS was 75.1 at% Ni and 24.9 at% Pt, which is similar to the TEM

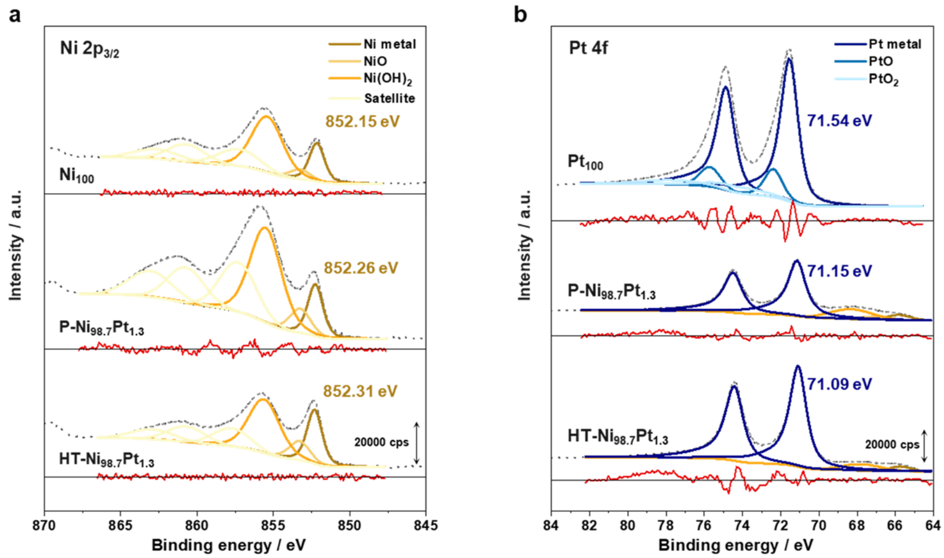

Fig. 2 XPS spectra of (a) Ni $2p_{3/2}$ and (b) Pt 4f region for Ni_{100}, Pt_{100}, P–$Ni_{98.7}Pt_{1.3}$, HT-$Ni_{98.7}Pt_{1.3}$. The raw intensity data are represented with a dotted line, and the obtained fitting data are presented with a gray short dash. The fitting residuals are presented at the bottom of each spectrum.

mapping results (79.8 at% Ni and 20.2 at% Pt). This Pt enrichment reflects surface energy minimization, consistent with DFT calculations below. After annealing at 200 °C, extensive surface reconstruction occurred, yielding a dramatic compositional shift to 49.2 at% Ni and 50.8 at% Pt, in relatively good agreement with the composition obtained from TEM elemental mapping (57.2 at% Ni and 42.8 at% Pt), with a minor (~8%) difference due to the difference in probing depth. Together with electron transfer from Ni to Pt, this thermally driven enhancement of surface Pt content further establishes a more electronically favorable and structurally stable catalytic interface, which is expected to play a critical role in catalytic performance.

To gain a fundamental understanding of the observed surface reconstruction, we modeled the atomic structure of Ni_3Pt_1 based on the experimentally determined surface composition (75.1/24.9 by XPS) (Fig. 3a and b). Considering the possibility of Pt surface segregation upon thermal annealing, we calculated the surface energies and cohesive energies as a function of Pt surface coverage (Fig. 3c and d). As shown in Fig. 3c, the surface energy at 25% Pt coverage was 0.113 eV Å^{-2}, which decreased to 0.100 eV Å^{-2} at 50% coverage, indicating enhanced surface stabilization. While further Pt enrichment slightly increased the surface energy (0.102 eV Å^{-2} at 75% and 0.110 eV Å^{-2} at 100% coverage), all values remained lower than that of the pure Pt (0.123 eV Å^{-2}). These results confirm that HT-$Ni_{98.7}Pt_{1.3}$ attained a thermodynamically stabilized surface structure with low surface energy, specifically at the $Ni_{49}Pt_{51}$ composition derived from XPS, through thermally induced surface reconstruction.

Cohesive energy calculations (Fig. 3d) further highlighted the structural stabilization of the Ni–Pt surface alloy. At Pt coverages of 25–50%, the cohesive energy was lower than that of pure Pt, indicating that Ni–Pt interactions provide a more favorable bonding environment compared to Pt–Pt interactions. This demonstrates that the formation of a Ni–Pt surface alloy not only reduces surface energy but also strengthens atomic bonding within the structure. Taken together, the reductions in surface and cohesive energies indicate that the HT-$Ni_{98.7}Pt_{1.3}$ surface is thermodynamically and structurally stabilized. These properties are expected to impart strong durability during electrochemical testing conditions.

Electrochemical measurements for HER activity and stability

The catalytic activity of the Ni, Pt, and Ni–Pt catalysts toward acidic HER was evaluated in 0.5 M H_2SO_4 (Fig. 4a–d). The polarization curves in Fig. 4a demonstrate that the Ni_{100} catalyst exhibited very limited activity, requiring a high overpotential of ~130 mV to reach a current density of −10 mA cm^{-2}. Strikingly, alloying Ni with 1.3 at% of Pt ($Ni_{98.7}Pt_{1.3}$) decreased the overpotential to 12 mV. When the bulk Pt composition was varied between 0.2 and 1.3 at%, the overpotential remained nearly consistent (Fig. S4), indicating that even trace amounts of Pt can markedly enhance performance. Furthermore, after heat treatment, the overpotential of HT-$Ni_{98.7}Pt_{1.3}$ remained at 14 mV, showing negligible performance loss. In addition, the performance of these Ni–Pt catalysts was consistently superior to that of pure Pt_{100}, which required an overpotential of 20 mV at −10 mA cm^{-2}.

The overpotentials required to achieve higher current densities were also evaluated (Fig. 4b). At −50 mA cm^{-2}, Ni_{100} required 210 mV and Pt_{100} required 77 mV. In contrast, $Ni_{98.7}Pt_{1.3}$ and HT-$Ni_{98.7}Pt_{1.3}$ required only 44 mV, confirming their superior performance at elevated current densities, which is crucial for practical HER applications. To verify the cost-effectiveness of the developed HT-$Ni_{98.7}Pt_{1.3}$ catalyst, the Pt mass activities of HT-$Ni_{98.7}Pt_{1.3}$ (Pt loading of 0.029 mg cm^{-2}) and Pt_{100} (Pt loading of 0.077 mg cm^{-2}) were compared at an

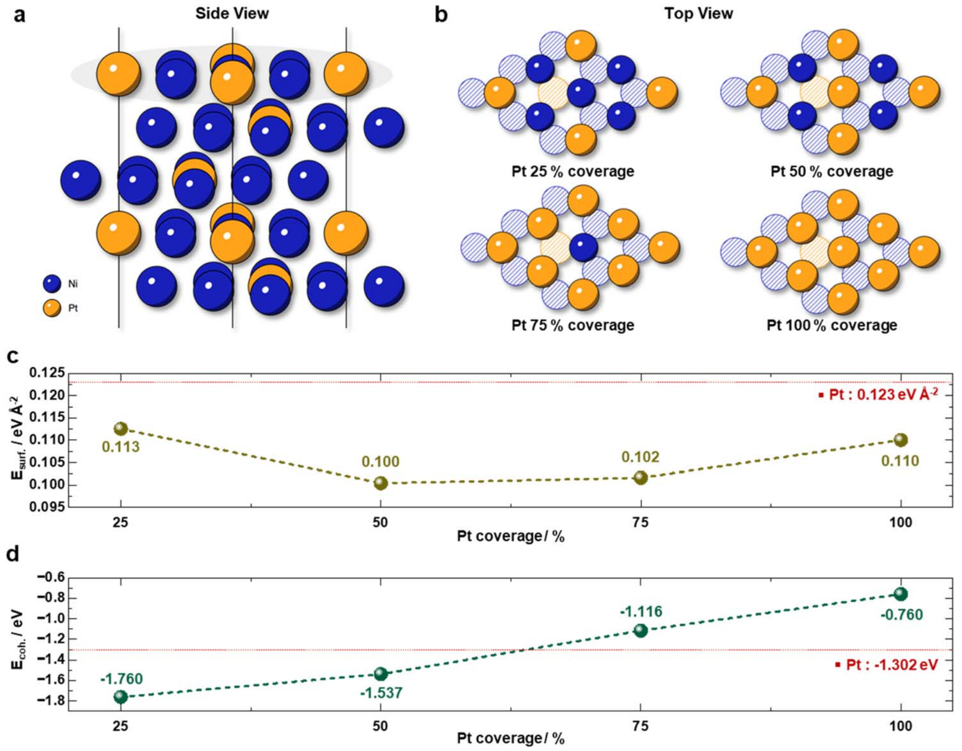

Fig. 3 DFT calculations. (a) Side view and (b) top view of the Ni_3Pt_1 model. (c) Surface energy and (d) cohesive energy according to Pt coverage.

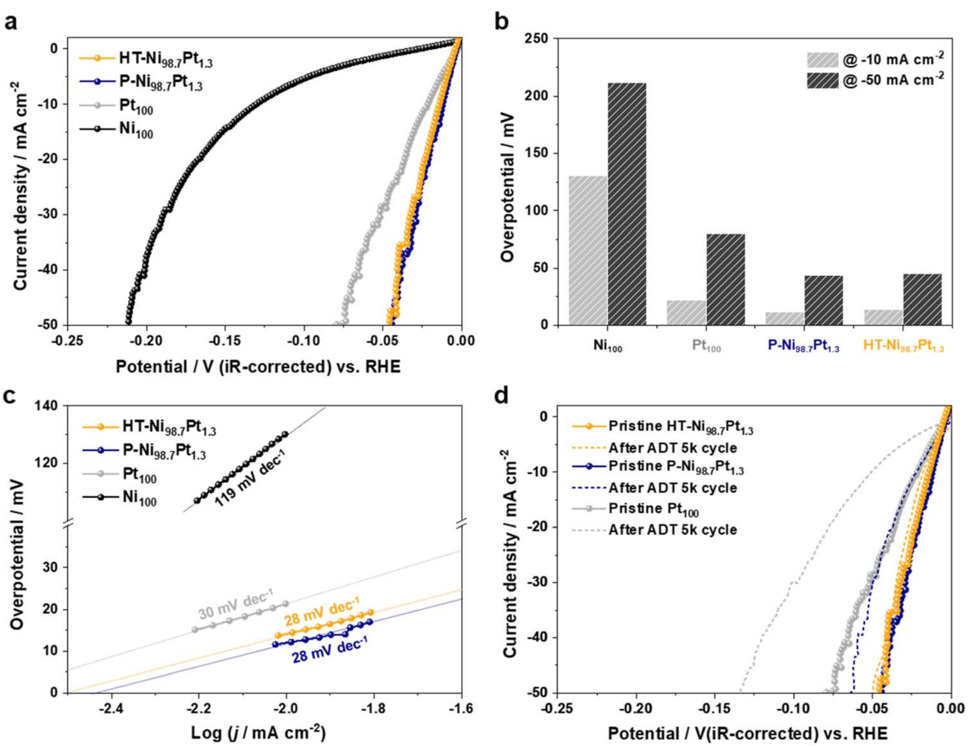

Fig. 4 (a) HER polarization curves of Ni_{100}, Pt_{100}, P–$Ni_{98.7}Pt_{1.3}$, and HT-$Ni_{98.7}Pt_{1.3}$ samples. (b) Overpotential at −10 and −50 mA cm^{-2}. (c) Tafel plots derived from the HER polarization curves of the corresponding catalysts. (d) Accelerated degradation test (ADT) for 5000 cycles at a scan rate of 100 mV s^{-1}.

overpotential of 50 mV (Fig. S5). HT-$Ni_{98.7}Pt_{1.3}$ exhibited 5.39 times higher mass activity than Pt_{100}, demonstrating superior Pt utilization efficiency and improved cost-effectiveness.

To gain mechanistic insight, Tafel plots were derived from the polarization curves (Fig. 4c). The Ni_{100} catalyst exhibited a Tafel slope of 119 mV dec^{-1}. A Tafel slope in this range is often interpreted as indicative of proton adsorption (Volmer step) being the rate-limiting step, corresponding to a theoretical value of 118 mV dec^{-1}. However, considering the complex dependence of the slope on surface hydrogen coverage, high slopes in this range can also originate from a Heyrovsky-limiting.[44] Because direct information regarding hydrogen adsorption behavior is not available for the present Ni catalyst, both Volmer- and Heyrovsky-controlled kinetics remain viable interpretations. The incorporation of 1.3 at% Pt into Ni (P–$Ni_{98.7}Pt_{1.3}$) caused the Tafel slope to decrease sharply from 119 mV dec^{-1} (for Ni_{100}) to 28 mV dec^{-1}. Even after heat treatment (HT-$Ni_{98.7}Pt_{1.3}$), the Tafel slope remained unchanged at 28 mV dec^{-1}, consistent with the theoretical value associated with a Tafel-controlled pathway (28 mV dec^{-1}), in which two adsorbed protons combine to form H_2. The pronounced decrease in Tafel slope suggests a cooperative interaction between Pt and Ni upon alloying. For Pt_{100}, the Tafel slope was measured to be 32 mV dec^{-1}, also close to the theoretical value. Nevertheless, the lower Tafel slope of the Ni–Pt catalyst underscores its superior HER kinetics compared to pure Pt.

To identify the origin of the enhanced catalytic activity, we performed DFT calculations to evaluate the hydrogen adsorption energy (ΔE_{H^*}). Using the previously constructed Pt alloyed Ni surface models (Fig. 3), we calculated the adsorption energies of hydrogen at four distinct hollow sites, where hydrogen is known to preferentially adsorb on the (111) surface of fcc metals (Fig. 5a–d). The average ΔE_{H^*} values for the four sites are summarized in Fig. 5e. On pure Pt, the average ΔE_{H^*} at the hcp-hollow site (−0.339 eV) and fcc-hollow site (−0.381 eV) was −0.360 eV, exceeding the optimal value (−0.24 eV) for HER.[44] It is widely accepted that slightly weaker ΔE_{H^*} than that on pure Pt surface can facilitates HER,[44] whereas excessively weak binding results in insufficient surface coverage of H* species, thereby limiting the overall reaction.

At 25% Pt coverage, the average ΔE_{H^*} was −0.459 eV, indicating stronger binding than that on pure Pt. However, with increasing Pt coverage, the adsorption strength progressively decreased, approaching 0 eV at 100% coverage. Considering the contributions from zero-point energy and the entropy of adsorbed H (∼0.24 eV),[44] the adsorption free energies at 75% and 100% Pt coverage became too weak to sustain adequate H* surface coverage. At 50% Pt coverage, the adsorption energy was −0.335 eV, slightly weaker than that of the pure Pt surface, which can facilitate the surface diffusion of H* and enhance HER activity. Collectively, these results reveal that alloying an optimal amount of Pt (*e.g.*, 50% coverage) onto the Ni surface provides a more favorable binding strength for hydrogen intermediates, thereby rationalizing the superior HER activity of HT-$Ni_{98.7}Pt_{1.3}$. This computational insight, alongside the experimentally verified surface reconstruction and stabilization, confirms that the unique Ni–Pt surface alloy structure is highly advantageous for hydrogen evolution. Although the averaged H-adsorption energies suggest optimal activity at intermediate Pt coverage, the relatively high HER performance

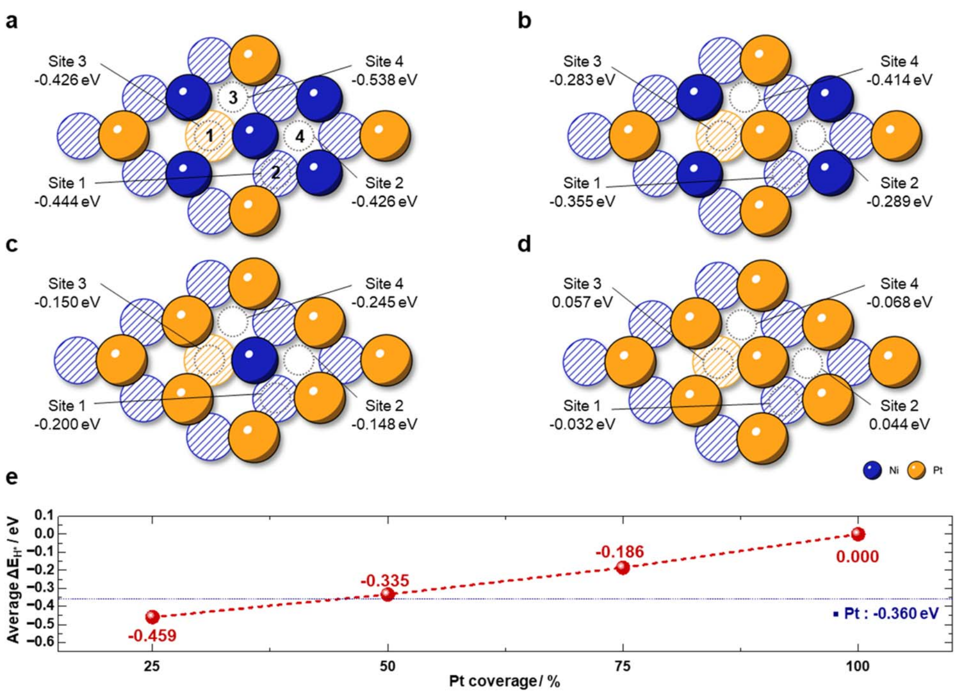

Fig. 5 Schematic of the Ni_3Pt_1 atomic structure and the corresponding H adsorption energies (ΔE_{H^*}) at various sites: (a) Pt 25% coverage, (b) Pt 50% coverage, (c) Pt 75% coverage, (d) Pt 100% coverage. (e) Average of the ΔE_{H^*} as a function of Pt coverage.

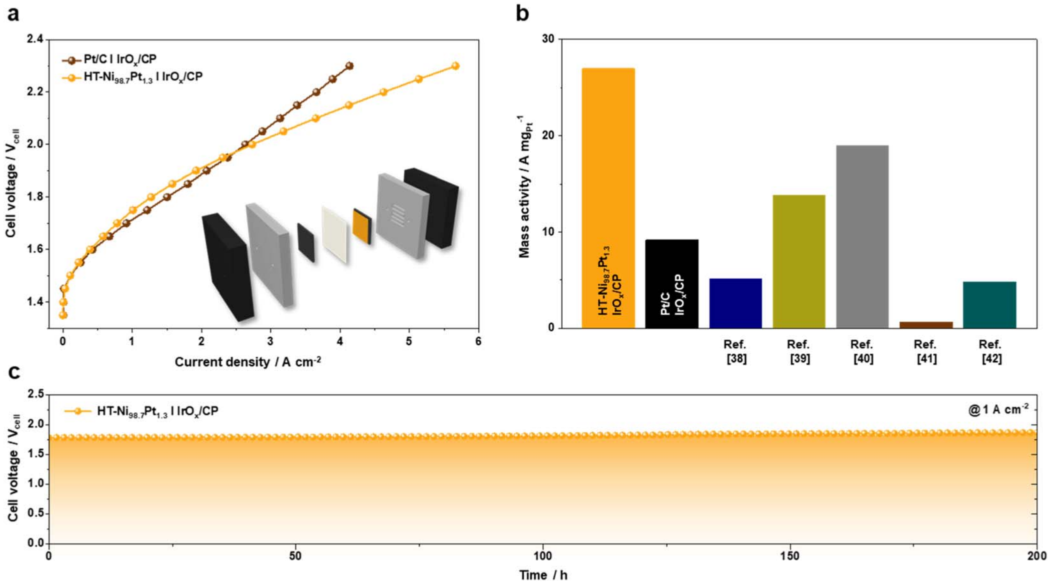

Fig. 6 (a) Single-cell polarization curves using HT-$Ni_{98.7}Pt_{1.3}$ (0.029 mg of Pt per cm^2)/CP cathode and IrO_x/CP anode. A commercial Pt/C cathode (0.1 mg of Pt per cm^2) based PEMWE performance is compared; inset image shows a schematic of the PEM electrolyzer device. (b) Performance comparison of the HT-$Ni_{98.7}Pt_{1.3}$ cathode integrated PEMWE cell with those reported previously based on current density and mass activity at 1.7 V_{cell}. (c) Long-term stability of the cell with HT-$Ni_{98.7}Pt_{1.3}$ cathode investigated at a current density of 1 A cm^{-2} for 200 h.

at 25% coverage may arise from local structural and electronic effects. At low coverage, Pt atoms could be atomically dispersed or form small ensembles on Ni, creating sites with near-optimal ΔG_{H^*}. The as-deposited 25% Pt layer may therefore contain structurally heterogeneous domains, whereas the annealed 50% Pt coverage surface becomes more ordered and stable. Consequently, the DFT model based on 50% coverage is likely closer to the experimental surface, while the 25% case may involve multiple metastable configurations. Overall, the observed HER behavior likely reflects the combined influence of structural diversity and kinetics rather than the averaged adsorption energy alone. Nevertheless, DFT inherently cannot fully capture the diverse surface structures possible at low Pt coverage.

Catalyst stability was assessed *via* ADT, consisting of 5000 cycles at a scan rate of 100 mV s^{-1} within the same HER evaluation potential range (Fig. 4d). Following ADT, Pt_{100} exhibited a pronounced increase in overpotential by 28 mV (from 22 to 50 mV) to reach −10 mA cm^{-2}, primarily attributed to the aggregation of Pt particles (Fig. S6c *vs.* S2f). P–$Ni_{98.7}Pt_{1.3}$ also showed a moderate overpotential increase of approximately 10 mV (from 12 mV to 22 mV) after ADT. This performance decline appears to stem from a combined effect of particle aggregation and partial dissolution from the Ni-rich phase. Indeed, SEM images recorded post-ADT (Fig. S6a *vs.* Fig. 1b) suggest significant catalyst material dissolution in P–$Ni_{98.7}Pt_{1.3}$. In contrast, HT-$Ni_{98.7}Pt_{1.3}$ demonstrated exceptional stability, with an overpotential change of only ∼3 mV after ADT. The remarkable durability of HT-$Ni_{98.7}Pt_{1.3}$, together with the observed reduction in both dissolution and aggregation observed in SEM (Fig. S6b *vs.* Fig. 1c), strongly correlates with computational prediction of low surface energy and robust Pt–Ni interactions on the optimized surface composition (Fig. 3).

Application and evaluation in PEMWE single cell

To validate the outstanding HER performance observed in half-cell tests, the HT-$Ni_{98.7}Pt_{1.3}$ electrode selected from the half-cell screening was integrated as the cathode in a full PEM water electrolysis single cell operated at 90 °C over a voltage window of 2.3–1.35 V_{cell} (Fig. 6a). A spray-deposited IrO_x on either CP or Ti was employed as the anode, with a Nafion 212 membrane positioned between the electrodes to form the MEA. The inset image in Fig. 6a presents the schematic of this PEMWE system, illustrating the integrated HT-$Ni_{98.7}Pt_{1.3}$/CP cathode and the IrO_x anode.

The PEMWE equipped with the HT-$Ni_{98.7}Pt_{1.3}$/CP cathode together with an IrO_x/CP anode delivered high performance, achieving a current density of 2.7 A cm^{-2} at a cell voltage of 2.0 V_{cell} (Fig. 6a). This performance is comparable to that of a PEMWE employing a commercial Pt/C/CP cathode (0.1 mg_{Pt} cm^{-2}) and IrO_x/CP anode, which reached 2.65 A cm^{-2}. Notably, the PEMWE with HT-$Ni_{98.7}Pt_{1.3}$/CP cathode outperformed the counterpart using a Pt/C/CP cathode at a voltage above 2 V_{cell}, delivering a current density of 5.6 A cm^{-2} at 2.3 V_{cell}, approximately 1.33 times higher than that of Pt/C (4.2 A cm^{-2}), despite the Pt loading being more than three times lower. This superior performance at high voltage is attributed to the inherent advantages of the catalyst-integrated electrode architecture, which significantly reduces mass transport resistance compared to conventional powder-type catalysts.[35,45] The superior performance of the HT-$Ni_{98.7}Pt_{1.3}$/CP cathode over the conventional Pt/C cathode was maintained even when IrO_x was applied onto a Ti fiber felt substrate to enhance corrosion resistance. Although PEMWE employing the HT-$Ni_{98.7}Pt_{1.3}$/CP cathode with an IrO_x/Ti anode exhibited a performance decrease compared to that with an IrO_x/CP anode, due to the higher resistance of Ti (Fig. S7), it delivered superior performance relative to the system using a Pt/C cathode with an IrO_x/Ti anode.

A comparative analysis of the PEWME performance with some representative values reported in the literature at 1.7 V_{cell} and 2.0 V_{cell} is presented in Table S3, including detailed conditions such as electrode compositions, membrane type, and testing temperature. At 2 V_{cell}, the performance of PEMWE equipped with the HT-$Ni_{98.7}Pt_{1.3}$/CP cathode and an IrO_x/CP anode (2.7 A cm^{-2}) is on par with or exceeds that of other low-Pt cathodes reported (Table S3, 1.340–2.633 A cm^{-2} at 2 V_{cell}).[46–50] In terms of performance per Pt mass, our PEMWE cell achieved an impressive 27 A mg_{Pt}^{-1} at 1.7 V_{cell} (Fig. 6b), approximately three times higher than commercial Pt/C cathodes and 2–40 times greater than other reported low-Pt cathodes, highlighting the exceptional efficiency and practical utility.

Operational stability of the electrode was evaluated by continuously tracking the cell voltage under a current density of 1 A cm^{-2} over 200 h (Fig. 6c). The PEMWE with HT-$Ni_{98.7}Pt_{1.3}$/CP cathode maintained stable performance throughout the test, corresponding to an average degradation rate of 0.43 mV h^{-1}. To rigorously monitor the leaching behavior of Ni and Pt, ICP-MS analysis was performed on the effluent collected after the 200 h stability test. Even after 200 h of continuous single-cell operation under harsh acidic PEMWE conditions, the cumulative metal leaching remained very low, with Ni and Pt accounting for only 5.11% and 6.13% of their initial contents, respectively. These results provide a direct quantitative measure of the chemical stability of the developed cathode under operating conditions, indicating minimal material dissolution during long-term electrolysis. Taken together, consistent results obtained from both half-cell HER measurements and full single-cell operation highlight the intrinsic catalytic activity and operational stability of the HT-$Ni_{98.7}Pt_{1.3}$ electrode under relevant electrochemical conditions, rather than solely reflecting overall device performance. These findings support its potential as an efficient and durable low-Pt cathode for future PEM water electrolysis systems.

Conclusions

We developed a Ni–Pt surface alloy catalyst through electrodeposition followed by thermal annealing, achieving a synergistic improvement in catalytic activity and structural stability. Thermal treatment induced surface segregation of Pt atoms, resulting in a Pt-enriched shell over a Ni-rich core—effectively generating a core–shell-like surface alloy structure. This surface reconstruction significantly reduced both surface and cohesive energies relative to pure Pt, indicating improved thermodynamic and structural stability. DFT calculations revealed near-optimal hydrogen at approximately 50% Pt surface coverage. These findings provide a clear mechanistic rationale for the experimentally observed enhancement in HER activity. The heat-treated $Ni_{98.7}Pt_{1.3}$ electrode exhibited a very low overpotential of 14 mV at −10 mA cm^{-2} in a half-cell HER test and demonstrated high mass activity (27 A mg_{Pt}^{-1} at 1.7 V_{cell}) along with remarkable stability over 200 h under 1 A cm^{-2} PEMWE single-cell operation. This work highlights the potential of surface alloy engineering guided by both experimental validation and computational insights as a powerful strategy to develop efficient, durable, and economically viable HER catalysts for scalable hydrogen production.

Author contributions

K.-R. Yeo: conceptualization, experimental investigation and data measurements, funding acquisition, writing – original draft. D. Kim: computational model and simulations, writing – original draft. H. Kim: experimental investigation and data measurements. S. Yoo: conceptualization, writing – review & editing. J. H. Jang: conceptualization, writing – review & editing. H. Park: computational model and simulations, supervision, writing – review & editing. S.-K. Kim: conceptualization, supervision, funding acquisition and project administration, writing – original draft, writing – review & editing.

Conflicts of interest

There are no conflicts to declare.

Data availability

The data supporting this article have been included as part of the supplementary information (SI). Supplementary information: Fig. S1–S7 and Tables S1–S3. See DOI: https://doi.org/10.1039/d6ta01806h.

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